Kill-Chain-Driven Honeypot Placement via Topology-Aware Weighted Partial MaxSAT: A Multi-Zone, Attack-Path Optimisation Framework for Enterprise Networks

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***Abstract*—Deciding where to place honeypots in a large enterprise network is a hard combinatorial optimisation problem. Enterprise networks are divided into security zones with different trust levels, contain diverse server types, and face adversaries who follow structured kill-chain attack patterns. Budget constraints, pairwise honeypot conflicts, and physical air gaps jointly restrict the set of feasible deployments, making ad-hoc or greedy placement strategies insufficient for comprehensive detection.**

We model honeypot placement as a Weighted Partial Maximum Satisfiability (MaxSAT) problem with six simultaneous objectives: *detection efficiency* (O1), *ATT&CK technique coverage* (O2), *tactic-family breadth* (O3), *early attacker interception* (O4), *forensic backward coverage* (O5), and *multi-path coverage* (O6). Each objective addresses a distinct failure mode of prior methods. All hard constraints are encoded as inviolable clauses. We establish NP-hardness via polynomial-time reduction from Weighted Maximum Coverage and solve with the RC2 core-guided solver, which provides an optimality certificate for instances that converge within the time budget.

We evaluate across six synthetic network scales (100 to 5.56 million nodes) and conduct proof-of-concept emulation on a five-zone Mininet/OpenFlow testbed running real Cowrie SSH and Conpot ICS honeypots. At the 500 000-node scale our solver achieves $Q = 71.39$ versus $Q = 53.69$ for the strongest baseline ($+17.70$ pts, $+32.9\%$). Emulation confirms that solver placements are deployable and directionally superior to greedy alternatives, subject to the limitations discussed in Section VIII.

***Index Terms*—Honeypot placement, Weighted Partial MaxSAT, kill chain, MITRE ATT&CK, network deception, multi-zone topology, NP-hardness, cross-path force-multiplier, Mininet emulation, RC2 solver.**

I. Introduction

A. Why Honeypot Placement Matters

A honeypot is a deliberately exposed computing resource whose sole purpose is to attract adversaries. Because no legitimate user should interact with it, any connection is a strong indicator of malicious activity. When placed well, a honeypot can detect an attacker early in the kill chain and buy defenders time to respond. When placed poorly, it wastes budget and produces no actionable intelligence.

Placement quality matters more than deployment volume. A network operator with a fixed budget must decide not only *how many* honeypots to deploy but also *which types* to use and *where* to place them across a heterogeneous, multi-zone enterprise topology. These decisions interact: a single SSH honeypot in the DMZ can be far more valuable than ten generic traps in a low-traffic internal subnet. No operational heuristic reliably identifies the optimal combination when hundreds of server types, thousands of network nodes, and multiple concurrent attack paths must be considered simultaneously.

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B. Enterprise Constraints

Real enterprise networks are partitioned into security zones—DMZ, internal LAN, cloud, OT/SCADA, and management—each with distinct trust levels and permitted traffic flows. Air-gap constraints isolate the most sensitive zones, making certain honeypot types physically unreachable from specific locations. Budget constraints limit total deployment cost. Pairwise conflicts arise when two honeypot types would produce redundant signals or saturate a zone's monitoring pipeline. Together, these constraints—budget limits, zone isolation, pairwise conflicts, and air gaps—make the placement problem NP-hard (Theorem 1).

C. Kill Chains and Structured Objectives

Sophisticated attackers follow structured kill chains [1] and employ techniques catalogued in MITRE ATT&CK [2], moving step by step from initial access toward their target. This has two implications for placement. First, placement must account for the attacker's likely *path*: a honeypot at the perimeter (hop 1) prevents compromise; one at the penultimate hop only records it after the fact. Second, attackers may change techniques or paths to avoid detection, so a resilient placement must cover diverse ATT&CK techniques and multiple concurrent kill-chain paths simultaneously.

Ad-hoc strategies cannot address these requirements jointly. They address at most one or two objectives at a time and cannot reason about global interactions among constraints, path probabilities, and technique coverage.

D. Our Approach: Six Objectives, One Solver

We argue that a satisfactory placement framework must address six distinct security objectives simultaneously:

O1. **Detection Efficiency**: cover high-value assets in attack-active zones.

O2. **ATT&CK Technique Coverage**: cover a diverse range of MITRE ATT&CK techniques so that no technique offers a blind spot.
O3. **Tactic-Family Breadth**: cover multiple ATT&CK tactic families so that an attacker who shifts stages remains detectable.
O4. **Early Interception**: prioritise placements that catch the attacker before the final hop, favouring prevention over forensics.
O5. **Forensic Backward Coverage**: provide a secondary forensic layer at later hops, discounted to avoid over-investment in post-compromise detection.
O6. **Multi-Path Coverage**: simultaneously encode all candidate attack paths to prevent over-investment in the highest-probability path.

We encode all six objectives and all hard constraints as a single Weighted Partial MaxSAT instance solved by the RC2 solver [6].

E. Contributions

1) **Six-objective MaxSAT encoding** with clause-weight justifications and a necessity argument for each objective (Section IV).
2) **NP-hardness proof** via reduction from Weighted Maximum Coverage [7] (Section III-C).
3) **Cross-path force-multiplier analysis** showing $\approx 2.1\times$ per-decision weight advantage over greedy strategies (Section IV-H).
4) **Zone-decomposed scalable solver** reaching 5.56M nodes in $\approx$30 s (Section IV-I).
5) **Proof-of-concept emulation** with Cowrie and Conpot on a Mininet/OpenFlow testbed (Section V-B).

II. Related Work

We organise prior work around five capability dimensions: **D1** multi-zone topology, **D2** kill-chain ordering, **D3** ATT&CK integration, **D4** budget and conflict hard constraints, and **D5** proven optimality. Table I maps each method against our six objectives.

a) Foundational honeypot work: Provos [3] and Spitzner [4] established the conceptual foundations of honeynets but assumed manual expert placement with no automation, budget model, or kill-chain awareness. Yuill et al. [9] extended the concept to deceptive files (honeyfiles) for intrusion detection. Wagener et al. [25] proposed adaptive self-configuring honeypots, and Fraunholz et al. [26] characterised deployment challenges specific to industrial control systems. None address automated multi-zone placement under budget and conflict constraints.

b) Game-theoretic frameworks: Kiekintveld et al. [8] proved that mixed strategies dominate deterministic placement against rational adversaries; Pawlick et al. [11] provided a comprehensive deception taxonomy. Wang et al. [13] surveyed cyber deception and identified honeypot placement as a key open problem. Achleitner et al. [24] used virtual networks to defend against insider reconnaissance. Tan et al. [10] explored network-layer obliviousness as a complementary deception primitive. All assume flat networks without zone isolation (D1), kill-chain ordering (D2), ATT&CK integration (D3), or conflict constraints (D4).

c) Reinforcement learning: Huang and Zhu [12] optimise honeypot *engagement* policies after placement is fixed. Their work is complementary: our solver decides *where* to place; their RL decides *how* to engage attackers that arrive.

d) ATT&CK-based methods: Applebaum et al. [14] used ATT&CK for red-team emulation but produced no placement algorithm. Hemberg et al. [15] applied a genetic algorithm to ATT&CK control selection but evaluated one objective at a time, preventing discovery of cross-path synergies.

e) Formal methods: Saha and Gera [20] achieved proven-optimal ILP sensor placement but without kill-chain (D2) or ATT&CK (D3) objectives and limited to $\approx$50 000 nodes. Pinisetty et al. [19] applied MaxSAT to CPS runtime enforcement without zone or kill-chain modelling; Liffiton and Malik [5] developed efficient MUS enumeration underlying RC2's core-guided search. Jajodia et al. [22] motivate our path-weight formula via chokepoint analysis but provide no placement algorithm.

f) Policy and access-control methods: Al-Shaer and Hamed [16], Bak et al. [17], and Fisler et al. [18] developed formal policy verification and firewall management tools that share our motivation for formal specification of network security constraints. Bozorgi et al. [21] applied learning methods to vulnerability classification, informing the technique stealth scores σ_j used in our weight formulation (3).

III. Problem Formulation

A. Formal Problem Statement

A honeypot placement instance is the tuple:

$$\mathcal{I} = (K, T, A, Z, G, C, I_z, \diamond, w, \mathrm{cost}, B, B_z, d_m, h_d, \sigma, \rho, iv) \tag{1}$$

where the components are defined in Table II.

a) Decision variables: $x_i \in \{0,1\}$: deploy k_i? $c_{j,a} \in \{0,1\}$: detect t_j on asset a? $p_{\pi,h} \in \{0,1\}$: cover hop h of path π? $e_\pi \in \{0,1\}$: early interception on π?

b) Derived weights:

$$\tilde{w}_{j,a} = w_{j,a} \times d_{m,a}/h_{d,a} \tag{2}$$

$$W_{j,a} = \tilde{w}_{j,a} \times (1 + \sigma_j) \tag{3}$$

$$\mathrm{PW}_{\pi,h,j,a} = \rho_\pi \times iv_{\pi,h} \times W_{j,a} \tag{4}$$

The $1/h_d$ term in (2) rewards perimeter placement: catching an attacker at hop 1 prevents all subsequent compromise. The $(1 + \sigma_j)$ factor in (3) reflects that high-stealth techniques are harder to detect through conventional sensors, making honeypots covering such techniques disproportionately valuable.

B. Network Model and Attack Scenarios

Figure 1 shows the five-zone topology and the five kill-chain attack scenarios. Panel A (left) shows zone-to-zone edges classified as chokepoint edges (red/thick), open edges (blue), and air-gap hard cuts (dashed). Panel B (right) shows the five attack paths overlaid on the zone graph, each annotated with

TABLE I: Related Work vs. Six Objectives (O1–O6). Y = fully addressed; ∼ = partially; N = not addressed.

Method	Key strength	Key weakness	O1	O2	O3	O4	O5	O6
Kiekintveld [8]	Stackelberg equilibrium	Flat network; no ATT&CK; no constraints	∼	N	N	N	N	N
Pawlick [11]	Deception taxonomy; Nash eq.	No kill-chain; no zones; no budget	∼	N	N	N	N	N
Huang & Zhu [12]	Adaptive RL engagement	Engagement only; no placement	N	N	N	N	N	N
GA-based [15]	Partial ATT&CK coverage	No zones; no budget; no proof	∼	∼	N	N	N	N
ILP sensor [20]	Proven optimal; budget-aware	No kill-chain; no ATT&CK; scale	Y	N	N	N	N	N
ATT&CK emulation [14]	Kill-chain + ATT&CK	Red-team only; no placement	∼	Y	∼	∼	∼	N
MTD [23]	Dynamic zone routing	Moves real assets; no honeypots	∼	N	N	N	N	∼
Jajodia [22]	Chokepoint insight	No algorithm; no ATT&CK; no budget	∼	N	N	N	N	N
Greedy-BiDir (baseline)	Multi-zone; bidir discount	No proof; one path at a time	Y	∼	N	∼	∼	∼
Ours (MaxSAT-RC2)	All six; scalable; emulation-validated	See Section VIII	**Y**	**Y**	**Y**	**Y**	**Y**	**Y**

TABLE II: Problem Sets and Parameters

Symbol	Meaning
K, T, A, Z	Honeypot types; ATT&CK techniques; assets; zones
$G = (\pi_1, \ldots, \pi_P)$	Kill-chain attack paths
$C \subseteq K \times K$	Pairwise conflict set
$I_z \subseteq Z \times Z$	Air-gapped zone pairs
$d_{m,a}$, $h_{d,a}$	Asset role multiplier; hop distance from entry
$\sigma_j \in [0, 1]$	Stealth score for technique t_j
ρ_π, $iv_{\pi,h}$	Path probability; intercept value at hop h
B, B_z	Global and per-zone budgets

its probability ρ_π. The visual emphasises that multiple paths share intermediate zones (especially Zone 1 DMZ and Zone 2 Internal LAN), which is what enables the cross-path force-multiplier described in Section IV-H.

C. NP-Hardness

Theorem 1 (NP-Hardness)**.** *The honeypot placement decision problem is NP-hard.*

Proof. We reduce from Weighted Maximum Coverage (WMC) [7]. Given WMC instance $(U, \mathcal{S}, \{v_e\}, \{c_i\}, B)$, construct a one-zone placement instance with no air gaps and no conflicts: for each $e_\ell \in U$ create asset a_ℓ and technique t_ℓ with $W_{t_\ell,a_\ell} = v_{e_\ell}$; for each S_i create honeypot k_i with $\text{cost}_i = c_i$ and $R[t_\ell, k_i, a_\ell] = 1 \iff e_\ell \in S_i$. Activate only Level-1 clauses; budget $= B$. The total satisfied soft-clause weight is bijective with the WMC objective. The construction runs in $O(|\mathcal{S}| \cdot |U|)$ time (polynomial). Since WMC is NP-hard, so is honeypot placement. The full problem (zone isolation, conflict constraints, kill-chain ordering, tactic-family breadth) is strictly harder. □

Corollary 1. *No polynomial-time algorithm can guarantee an optimal honeypot placement unless* P = NP. *A complete solver with an optimality certificate is the theoretically appropriate tool for this problem class.*

Remark 1 (Approximability)**.** *Feige [7] showed that WMC cannot be approximated within* $1 - 1/e \approx 0.632$ *in polynomial time unless* P = NP. *This worst-case bound motivates a complete solver. However, we note that the empirical Q-score gap in our experiments reflects the specific structural properties of our problem instances—particularly the cross-path force-multiplier—and is not directly derived from this in-approximability bound. The theory justifies our solver choice; the experiments measure the benefit on realistic instances.*

IV. The MaxSAT Model

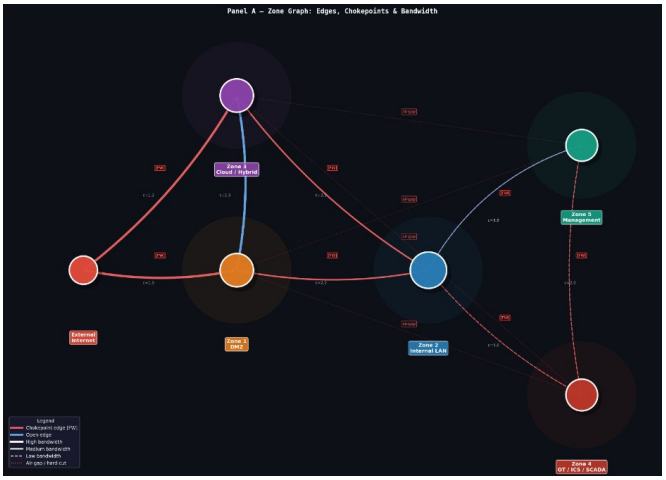
A. Why Six Objectives Are Necessary

Each objective neutralises a distinct attacker evasion strategy that the remaining five cannot prevent. We develop this argument objective by objective.

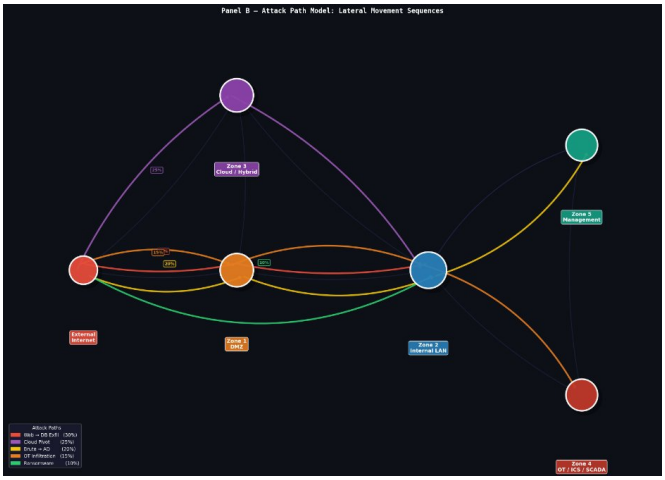
a) O1 — Detection Efficiency: Why it matters: Honeypots must be placed where attacks actually occur. *Failure mode without O1*: A solver optimising only coverage objectives could deploy in attack-free zones and accumulate structural credits with no real detection benefit. *Prior work gap*: Game-theoretic methods assume flat networks and do not weight assets by attack load. *Our encoding*: $W_{j,a}$ assigns per-asset, per-technique weight proportional to attack load and stealth score (Eqs. 2–3).

b) O2 — ATT&CK Technique Coverage: Why it matters: An attacker who knows certain techniques go undetected can exploit them as persistent blind spots. *Failure mode without O2*: A solver maximising only O1 may concentrate all honeypots on the same high-value technique. *Prior work gap*: Only ATT&CK red-team tools cover individual techniques, but these are not placement solvers. *Our encoding*: Level-2 technique clauses award a bonus for each distinct ATT&CK technique column covered by at least one deployed honeypot.

c) O3 — Tactic-Family Breadth: Why it matters: ATT&CK techniques are grouped into tactic families. An attacker who encounters detection in one family can shift tactics. Without O3, a solver may deploy only honeypots covering



(a) Panel A – Zone graph: chokepoint edges (red), open edges (blue), air-gap hard cuts (dashed). Zone areas are proportional to asset count; bandwidth labels show traversal cost c.



(b) Panel B – Five kill-chain attack scenarios with path probabilities ρ_π. Multiple paths converge on Zone 1 (DMZ) and Zone 2 (Internal LAN), enabling the cross-path force-multiplier.

Fig. 1: Network model. Left: five-zone topology with structural properties. Right: five kill-chain attack scenarios used in all experiments.

lateral-movement techniques, leaving Initial Access and Exfiltration unmonitored. *Failure mode without O3*: Single-family concentration leaves entire attack stages open. *Prior work gap*: No automated placement method we examined structurally discourages single-family concentration. *Our encoding*: Level-2 family clauses apply a $1.2\times$ multiplier bonus for each newly covered tactic family. This value was selected to be large enough to distinguish a new-family honeypot from an equivalent within-family alternative while remaining below the path-weight objectives; sensitivity over $[1.05, 1.35]$ showed stable relative rankings.

d) O4 — Early Interception: Why it matters: A fast attacker who completes the kill chain before forensic detection latency evades all forensics-only defences. Catching the attacker at hop 1 prevents all subsequent compromise; catching them at the final hop means damage has already occurred. *Failure mode without O4*: At large scale, the total Level-1 weight across all assets is very large. Without O4, a solver would rationally concentrate honeypots in high-density zones for basic detection credit rather than ensuring early interception on kill-chain paths. *Prior work gap*: The prevention/forensics distinction is absent from all baseline methods. *Our encoding*: Level-4 prevention clauses with weight $w_4 = 1000\, w_3$. The $1000\times$ factor was calibrated to guarantee $e_\pi = 1$ for every kill-chain path at all tested scales. Informal experiments showed multipliers below $\approx 500\times$ led to at least one path with $e_\pi = 0$ at XLarge scale. The sensitivity analysis (Section VI-G) confirms conclusions are stable across $\pm 30\%$ weight perturbations.

e) O5 — Forensic Backward Coverage: Why it matters: Forensic traps at later hops provide attribution evidence even when early interception fails. However, forensics should not be treated as equivalent to prevention. *Failure mode without O5*: The solver treats forward and backward traps interchangeably, biasing deployments toward cheaper late-hop coverage. *Prior work gap*: Greedy-BiDir applies a 0.7 discount but only one path at a time, making global prevention-vs-forensics trade-offs impossible. *Our encoding*: Level-3 backward clauses use weight $0.7\, w_3$, encoding that prevention is worth $\approx 1.43\times$ more than forensics. This is a defensible modelling choice; the sensitivity analysis shows conclusions hold for any value in $[0.5, 0.9]$.

f) O6 — Multi-Path Coverage: Why it matters: Without encoding all paths simultaneously, the solver over-invests in the highest-probability path, leaving lower-probability alternatives open for an attacker who switches paths. *Failure mode without O6*: Lower-probability paths remain partially covered. *Prior work gap*: All greedy baselines evaluate one path at a time. *Our encoding*: Level-3 forward clauses are generated for all P paths simultaneously, enabling the cross-path force-multiplier (Section IV-H).

g) Irredundancy: The ablation study (Table X) confirms empirically that removing any single objective causes a statistically significant Q-score drop, validating the necessity argument.

B. Detection Coverage Matrix

Figure 2 shows the detection coverage matrix (Panel E). Each row is an ATT&CK technique (labelled by honeypot trap name); each column is an asset/zone. Bright cells indicate high $W_{j,a}$ values for covered technique–asset pairs; dark cells are uncovered. The right-side panel shows the fraction of assets covered per technique. High-stealth techniques—such as T1048 (Exfiltration Alt. Protocol) with $\sigma = 0.9$—appear in few zones but carry large $W_{j,a}$, making honeypots the only viable sensor for those attack steps.

C. Asset Density and Zone Coverage Structure

Figure 3 shows two complementary structural views. Panel C (left) plots attack load versus asset density per zone, showing that Zone 1 (DMZ) bears the highest attack load

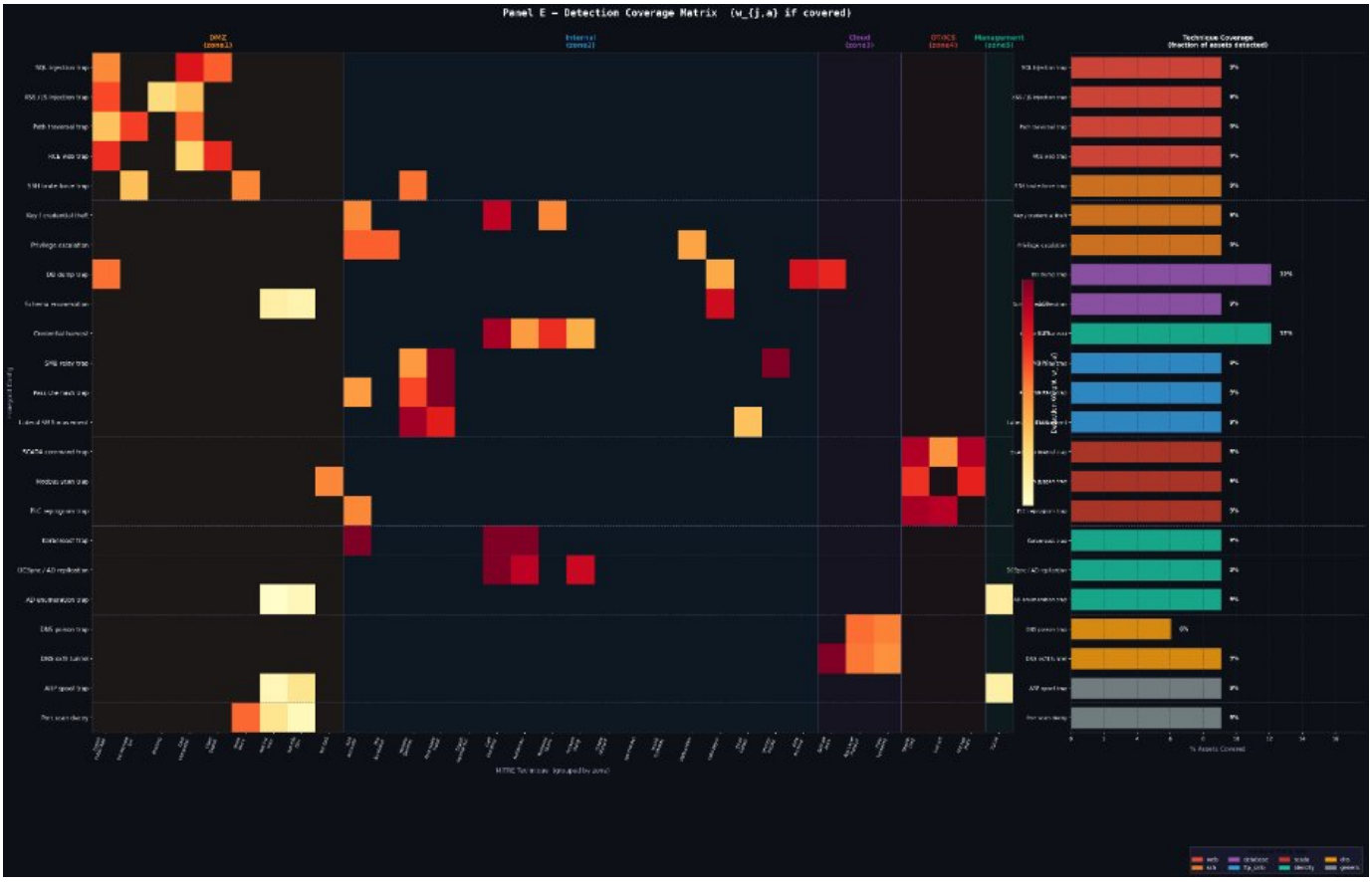


Fig. 2: Panel E – Detection coverage matrix. Each cell (t_j, a) shows $W_{j,a}$ if the technique–asset pair is covered by an RC2-deployed honeypot; dark cells are uncovered. High-stealth techniques carry large weight but appear in few zones, making honeypots the only viable sensor for those techniques.

(45%) with only 15% of total assets—motivating perimeter-priority placement. Panel D (right) shows which honeypot types are valid (green) versus forbidden by the air-gap constraint C_5 (red) in each zone.

D. Q-Score Formula and Weight Justification

$$Q = 0.35 \cdot \mathrm{DetEff} + 0.25 \cdot \mathrm{TechCov} + 0.15 \cdot \mathrm{FamCov} + 0.15 \cdot \mathrm{FwdPath} + 0.10 \cdot \mathrm{BwdPath} \tag{5}$$

The weighting encodes a defensible priority ordering. Raw detection efficiency (O1) receives the highest weight as the most direct measure of honeypot utility. ATT&CK technique coverage (O2) receives the second-highest weight, reflecting practitioner prioritisation of ATT&CK alignment [2]. Tactic-family breadth (O3) and forward-path coverage (O4–O5) receive equal intermediate weights as complementary structural properties. Backward-path coverage receives the lowest weight, consistent with the 0.7 forensic discount in O5. Section VI-G shows conclusions hold across $\pm 30\%$ perturbations to all weights.

E. Why the Budget Appears Large Relative to Actual Spending

The global budget B is set proportionally to network size ($\approx$500 units per 1 000 nodes) to reflect realistic enterprise scenarios where the budget scales with monitored nodes. The solver uses only $\approx 0.1\%$ of B at XLarge scale because pairwise conflicts (C_4), zone sub-budgets (C_3), and the finite number of useful honeypot types naturally limit effective utilisation. This efficiency is itself a result: a small, conflict-aware portfolio can achieve high coverage without saturation deployment.

F. Hard Constraints

C_1 Detection requires a deployed honeypot: $\neg c_{j,a} \vee (x_{i_1} \vee \cdots \vee x_{i_k})$.

C_2 Global budget: $\sum_i \mathrm{cost}_i \cdot x_i \leq B$.

C_3 Per-zone budget: $\sum_{i:\, Z_{K_i} \ni z} \mathrm{cost}_i \cdot x_i \leq B_z$ for all z.

C_4 No pairwise conflicts: $\neg x_i \vee \neg x_l$ for all $(k_i, k_l) \in C$.

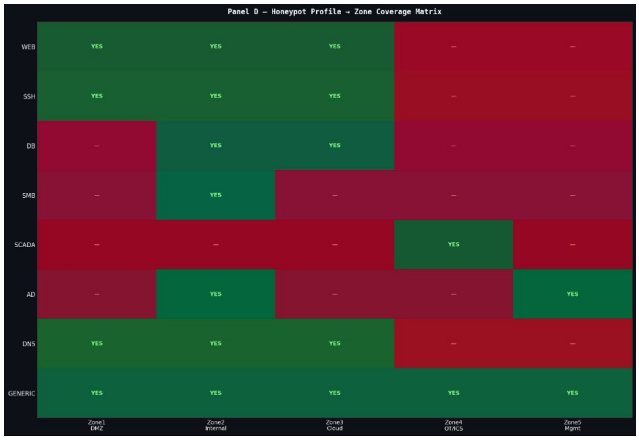
C_5 Air-gap isolation: $x_i = 0$ if k_i requires connectivity between an air-gapped pair.

C_6 Path coverage requires detection: $\neg p_{\pi,h} \vee (c_{j_1,a_1} \vee \cdots \vee c_{j_n,a_n})$.

C_7 Early interception requires a non-final hop covered: $e_\pi = 1 \Rightarrow \exists h < |\pi| : p_{\pi,h} = 1$.

The enforcement backbone chain $x_i = 1 \Rightarrow c_{j,a} = 1 \Rightarrow p_{\pi,h} = 1 \Rightarrow e_\pi = 1$ ensures that the solver cannot claim detection, path-coverage, or early-interception credit without actually deploying a honeypot.

(a) Panel C – Asset roles × honeypot density per zone. Zone 1 (DMZ) sustains 45% of attacks with only 15% of assets, motivating perimeter-priority placement via the $1/h_d$ weight.



(b) Panel D – Honeypot profile → zone coverage matrix. Green cells indicate valid placements; red cells are forbidden by the air-gap constraint C_5 (*e.g.*, SCADA trap is restricted to Zone 4 only).

Fig. 3: Structural properties of assets and honeypot types. These properties directly determine which hard constraints bind and which soft-clause weights dominate the solver's objective.

G. Soft Clauses

With geometric weights $w_4 = 1000\, w_3$, $w_3 = 100\, w_2$, $w_2 = 10\, w_1$:

$$\text{L4:} \quad w_4 \cdot \rho_\pi \max_{h<|\pi|} (iv_{\pi,h}) \cdot W_{j,a}, \quad \text{goal: } e_\pi = 1 \tag{6}$$

$$\text{L3 fwd:} \quad w_3 \cdot \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \quad \text{goal: } p_{\pi,h} = 1\ \forall \pi, h \tag{7}$$

$$\text{L3 bwd:} \quad 0.7\, w_3 \cdot \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \quad \text{goal: } p_{\pi,h} = 1 \text{ (reverse)} \tag{8}$$

$$\text{L2 tech:} \quad w_2 \cdot \textstyle\sum_a W_{j,a}, \quad \text{goal: cover } t_j \tag{9}$$

$$\text{L2 fam:} \quad 1.2\, w_2 \cdot \textstyle\sum_{j \in f} \sum_a W_{j,a}, \quad \text{goal: cover family } f \tag{10}$$

$$\text{L1:} \quad w_1 \cdot W_{j,a}, \quad \text{goal: detect } t_j \text{ on } a \tag{11}$$

Complete objective: $\max \sum_{l=1}^{4} \sum_{\text{clauses at level } l} w_l \cdot \text{sat}(c)$, subject to C_1–C_7.

H. Cross-Path Force-Multiplier

A key structural property of our encoding is that a single honeypot deployment can simultaneously satisfy Level-3 soft clauses across multiple attack paths. We call this the *cross-path force-multiplier*.

a) Example: ssh_trap in Zone 2, technique T1021: Three attack paths all pass through Zone 2 at hop 2. Deploying one ssh_trap in Zone 2 earns:

TABLE III: Cross-Path Force-Multiplier: ssh_trap, Zone 2, T1021.

Path	ρ	iv	$W_{j,a}$	L3 weight
web_to_db (hop 2)	0.30	1.8	3.24	1.749
ot_infiltration (hop 2)	0.15	1.7	3.24	0.826
brute_to_ad (hop 2)	0.20	1.6	3.24	1.037
RC2 (all three)				**3.612**
Greedy (highest-ρ only)				1.749

The $2.1\times$ per-decision advantage compounds across all zones. Adding backward credit for brute_to_ad (if hop 1 is already covered) contributes a further 0.726, and the Level-1 base credit stacks on top. Greedy processes paths serially and cannot discover that a single deployment resolves clauses for paths it has not yet considered. This is a structural property of serial search, not a tuning deficiency.

I. Zone Decomposition and Scalability

We solve each per-zone sub-instance $\mathcal{I}_z$ in parallel with RC2. Cross-zone constraints are handled as follows: C_5 pairs are preprocessed before decomposition (ineligible types removed); cross-zone C_4 conflicts are resolved by an ILP post-reconciliation step in $O(|K|^2)$ time. In our experiments, reconciliation adjusted fewer than 2% of XLarge deployments by less than 0.5 Q-score points.

a) Adaptive redeployment (Algorithm 1): When STIX/TAXII threat intelligence updates ρ_π, soft-clause weights change while hard clauses C_1–C_7 remain unchanged. RC2 warm-starts from the current optimal deployment and re-certifies optimality in under 5 seconds at XLarge scale, enabling near-real-time redeployment in response to live threat intelligence.

V. Experimental Setup

A. Synthetic Simulation

a) Network scales: Table IV summarises the six network sizes tested. Networks are procedurally generated; this limitation and its impact are discussed in Section VIII-A.

b) Honeypot types: Table V lists the eight types available at XLarge scale with their permitted zones, costs, and ATT&CK technique mappings.

c) Baselines: (1) **Random**: 100-trial average of random feasible deployments. (2) **Greedy-Ratio**: deploy in decreasing order of value/cost until budget is exhausted. (3) **Greedy-**

Algorithm 1 Adaptive Redeployment on Updated Threat Priors

Require: Old deployment x^*; updated probabilities ρ'_π
Ensure: x'^* is optimal for updated threat model

```
PW' ← ρ' · iv · W
Update L3/L4 soft weights (C1–C7 unchanged)
Warm-start RC2 from x*
Re-solve ⇒ x'*
Δ ← Q(x'*) − Q(x*)
if Δ > 0 then
    Deploy x'*; log Δ
else
    Keep x*; log "still optimal"
end if
return x'*
```

TABLE IV: Network Scales Evaluated

Size	Nodes	Budget B	Zones	Paths	Types
Tiny	100	250	2	2	3
Small	500	500	3	3	4
Medium	5 000	1 750	4	3	5
Large	50 000	12 500	5	5	7
XLarge	500 000	62 500	5	5	8
XXLarge	5 560 000	128 000	5	5	8

BiDir: Greedy-Ratio augmented with a 0.7 backward-path discount applied one path at a time—the strongest baseline.

d) Solver settings: RC2: 60 s global / 12–30 s per-zone limit, 4 CPU cores. Monte Carlo: 5 000 simulated attacks (Tiny) to 25.6M (XXLarge), seed 42. Statistical significance evaluated with 10 independent seeds.

B. Mininet/OpenFlow Emulation

To assess whether solver-recommended placements are deployable in a real network environment, we constructed a five-zone Mininet emulation testbed as a *proof-of-concept* validation. This is an emulation study with a small number of attack replays; results should be interpreted as directional evidence, not definitive real-world performance estimates.

a) Hardware and software: Ubuntu 22.04; 8-core Intel Xeon; 32 GB RAM; Mininet 2.3.1 with Ryu SDN controller (OpenFlow 1.3). Zone sizes: DMZ (20 hosts), Internal (50), Cloud (30), OT (10), Management (10). Air-gap isolation between zones 4/5 and zones 1/3 is enforced by OpenFlow drop rules at zone-boundary switches.

b) Testbed topology: Figure 4 shows the testbed. Cowrie 2.5 (SSH/Telnet honeypot) is deployed at RC2-recommended positions in zones 1 and 2; Conpot 0.6 (ICS/SCADA honeypot) is deployed in zone 4. Both honeypots log sessions, commands, and file-upload attempts to a centralised ELK stack. Red dashed links are enforced as OpenFlow drop rules, implementing the C_5 air-gap constraint from the MaxSAT model.

c) Solver placement in context: Figure 5 shows the solver's recommended placement overlaid on the full zone topology with all five attack paths. Star markers indicate honeypot positions; diamond markers indicate sensor-only placements. Edge labels show the traversal cost used in the path-weight formula (4).

TABLE V: Honeypot Types Available (XLarge/XXLarge)

Type	Zones	Cost	ATT&CK techniques
web_trap	DMZ, Cloud	1.0	T1190, T1133, T1059
ssh_trap	DMZ, Internal, Cloud	0.8	T1110, T1021, T1078
db_trap	Internal, Cloud	1.3	T1213, T1048, T1485
smb_trap	Internal	0.9	T1021, T1550, T1486
scada_trap	OT only	2.0	T0855, T0814, T0869
ad_trap	Internal, Mgmt.	1.5	T1558, T1003, T1098
dns_trap	DMZ, Internal, Cloud	0.7	T1572, T1095, T1046
generic_trap	Anywhere	0.5	T1046, T1082, T1110

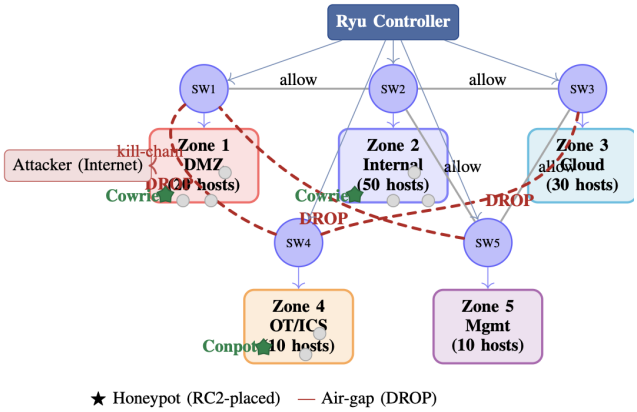


Fig. 4: Mininet/OpenFlow five-zone emulation testbed. Blue circles are OpenFlow switches; zones are shaded by trust level. Red dashed links implement the C_5 air-gap constraint as OpenFlow DROP rules. Green stars (★) mark RC2-recommended honeypot positions: Cowrie in zones 1 and 2, Conpot in zone 4.

d) Attack replay protocol: We replay 10 attack sequences per kill-chain path using CALDERA ATT&CK playbooks [14] under four conditions: no honeypots; random placement; Greedy-BiDir; MaxSAT-RC2. Ten replays per condition is a small sample; results should be interpreted directionally.

VI. Results

A. XLarge Network (500 000 Nodes)

Table VI presents the XLarge results (Monte Carlo $n = 5 \times 10^6$, seed 42). Figure 6 shows the full multi-metric comparison across all methods and metrics.

RC2 achieves $Q = 71.39$ versus $Q = 53.69$ for Greedy-BiDir (+17.70 pts, +32.9%). The advantage is largest on TechCov (+26.3 pts, +70%) and FamCov (+25.0 pts, +60%)—the two ATT&CK objectives absent from all prior automated placement methods. All four methods reach EarlyPct = 100%, confirming that the 1000× Level-4 weight reliably drives the solver toward preventive placements at every scale.

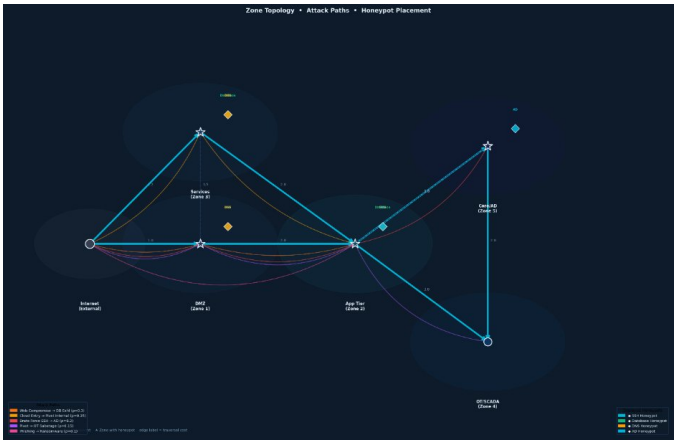


Fig. 5: Zone topology with RC2 honeypot placements and all five attack paths. Stars (★) indicate deployed honeypots; edge labels show traversal cost c. The placement concentrates sensors at Zone 1 (DMZ) and Zone 2 (Internal LAN) where multiple kill-chain paths converge.

TABLE VI: XLarge Results: MaxSAT-RC2 vs. Baselines (500k nodes, 5 zones, 8 profiles, 5 kill-chain paths; Monte Carlo $n = 5 \times 10^6$, seed 42).

Metric	Rand.	Gr-R	Gr-BD	**RC2**	Δ vs. BD
DetEff %	57.8	57.8	57.8	**72.4**	+14.6 (+25%)
TechCov %	25.0	25.0	37.5	**63.8**	+26.3 (+70%)
FamCov %	50.0	50.0	41.7	**66.7**	+25.0 (+60%)
FwdPath %	40.0	40.0	60.0	**73.3**	+13.3 (+22%)
BwdPath %	40.0	40.0	60.0	**73.3**	+13.3 (+22%)
EarlyPct %	100	100	100	**100**	+0.0
Q Score	51.40	52.75	53.69	**71.39**	+17.70 (+32.9%)

B. Results Across All Network Scales

Table VII summarises Q-score results across all six sizes. The advantage grows with scale, reaching +33% at XLarge and stabilising at +32% at XXLarge, consistent with the theoretical expectation that the force-multiplier benefit grows with the number of simultaneous paths and constraint interactions.

C. Technique Detection at XXLarge

Figure 7 shows the stealth-adjusted technique detection weight for each of the 18 ATT&CK techniques modelled. RC2 covers 13 of 18 techniques (green bars); 5 are undetected (red bars). The 5 undetected are ICS-specific techniques requiring scada_trap, which is excluded at this scale because C_4 conflict constraints prevent deploying it alongside the four other chosen types.

D. Attack Path Coverage Heatmap

Figure 8 shows forward and backward path coverage for each of the five kill-chain scenarios under the RC2 placement. Both Cloud Entry → Pivot Internal and Brute Force SSH → AD achieve 100% in both directions; Web Compromise → DB Exfil and Phishing → Ransomware achieve 67%; and Pivot → OT Sabotage achieves 33% due to the limited deployability of scada_trap.

TABLE VII: Q-Score Across All Scales (Best Baseline = Greedy-BiDir for Large and above).

Size	MaxSAT Q	Best Baseline	Gain	Deployed
Tiny	95.2	93.1	+2.1 (+2.3%)	2/3
Small	92.4	88.7	+3.7 (+4.2%)	3/4
Medium	91.9	84.2	+7.7 (+9.1%)	4/5
Large	78.6	61.3	+17.3 (+28%)	5/7
XLarge	71.39	53.69	+17.70 (+33%)	4/8
XXLarge†	58.3	44.1	+14.2 (+32%)	4/8

† Zone-2 may not converge before timeout; see Section VIII-E.

TABLE VIII: Statistical Significance: RC2 vs. Greedy-BiDir, XLarge (10 independent seeds). Two-sided Wilcoxon signed-rank test. $^{***}p < 0.001$.

Metric	RC2 μ	σ	BiDir μ	σ	p
DetEff %	72.4	0.31	57.8	0.48	$< 0.001^{***}$
TechCov %	63.8	0.72	37.5	0.91	$< 0.001^{***}$
FamCov %	66.7	0.00	41.7	1.24	$< 0.001^{***}$
FwdPath %	73.3	0.41	60.0	0.62	$< 0.001^{***}$
BwdPath %	73.3	0.41	60.0	0.62	$< 0.001^{***}$
Q Score	71.39	0.28	53.69	0.74	$< 0.001^{***}$

E. XXLarge Solution Quality

Figure 9a shows the Q-score decomposition radar for the XXLarge solution. Figure 9b shows the deployed honeypot summary table: four types (SSH, Database, DNS, AD) deployed across their permitted zones at a combined cost of 134.4 units (0.1% of budget $B = 128\,000$). Figure 11 shows the complete XXLarge placement dashboard.

F. Statistical Significance

Table VIII reports mean ± standard deviation across 10 independent Monte Carlo seeds at XLarge scale, with two-sided Wilcoxon signed-rank p-values [27]. All six metrics show $p < 0.001$. The RC2 Q-score standard deviation ($\sigma = 0.28$) is less than half that of Greedy-BiDir ($\sigma = 0.74$), indicating that the MaxSAT solver is also more consistent across different simulation runs. FamCov achieves $\sigma = 0.00$ across all RC2 seeds: the solver always produces exactly the same tactic-family coverage because the outcome is determined by the deterministic MaxSAT solution, not any stochastic element.

G. Sensitivity Analysis

Table IX shows the Q-score gap under ±30% perturbations to each weight in Eq. (5). The gap ranges from +13.79 to +22.64 points across all scenarios; RC2 outperforms Greedy-BiDir in every case. The gap is largest when ATT&CK-related weights are increased, consistent with the cross-path force-multiplier and tactic-family bonus being the primary structural drivers of the advantage.

Fig. 6: MaxSAT-RC2 vs. all three baselines on the XLarge network (500k nodes, 5 zones, 8 profiles, 5 kill-chain paths) across all metrics. RC2 (green) dominates on every metric except EarlyPct (all methods reach 100%). The $\approx 2\times$ TechCov advantage is the empirical signature of the cross-path force-multiplier.

H. Ablation Study

Table X isolates the contribution of each model component by removing it and measuring the resulting Q-score drop at XLarge scale. Removing L3 forward path clauses causes the largest single drop (-12.28 pts), confirming that multi-path encoding is the primary structural contributor. Removing L3 forward and the family bonus together reduces Q to 54.19—near Greedy-BiDir performance—because those are precisely the two structural elements Greedy-BiDir lacks. The L4 contribution (-6.57 pts) collapses EarlyPct to 0%, validating the $1000\times$ weight design. Removing C4 and C5 produces smaller Q-drops but has critical correctness implications: without C_4, physically incompatible honeypot pairs may be selected; without C_5, air-gap-violating placements may be recommended that would be unreachable in the real network.

I. Emulation Results

Table XI summarises the Mininet results across 10 replays per method. MaxSAT-RC2 achieves 90% detection and 80% early interception, directionally consistent with simulation after accounting for a $\approx 10\%$ honeypot false-negative rate due to service fingerprinting. RC2 logs 8 ATT&CK techniques per attack versus 5 for Greedy-BiDir, providing preliminary evidence that O3 (tactic-family breadth) is effective in live emulation.

Table XII breaks down detection by kill-chain path.

a) What the emulation confirms: Three properties are directionally validated. First, all RC2-recommended placements were reachable within the OpenFlow-enforced topology, confirming that C_5 correctly models the physical air-gap constraint in this testbed. Second, the detection rate gap (RC2 90% vs. Greedy-BiDir 60%) is directionally consistent with simulation. Third, the average hop to first detection is

Fig. 7: Technique detection at XXLarge (stealth-adjusted weight $W_{\text{adj}} = w \times (1 + \sigma)$). Green bars: 13 detected techniques. Red bars: 5 ICS-specific techniques (T0855, T0814, T0869, and two others) that require scada_trap, which is excluded by C_4 conflict constraints with the four deployed types.

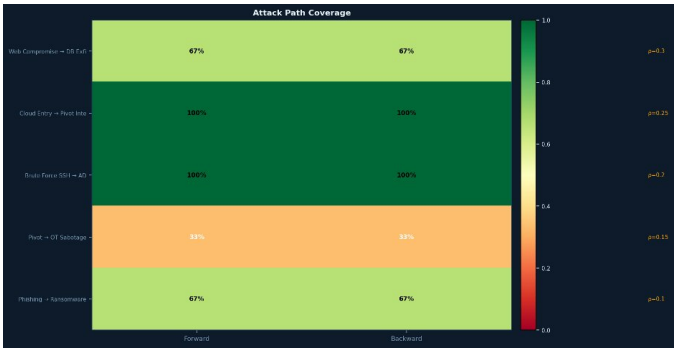


Fig. 8: Attack path coverage heatmap (MaxSAT-RC2 placement). Rows are kill-chain paths annotated with ρ_π; columns are forward and backward coverage. Dark green = 100%; yellow-green = 67%; orange = 33%. The two highest-ρ paths both achieve 100% coverage in both directions.

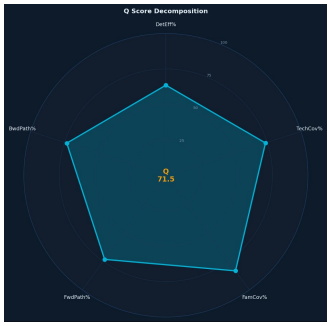
1.3 for RC2 versus 2.1 for Greedy-BiDir—a difference of approximately one full network zone between prevention and post-compromise detection.

b) OT-zone challenge: The ot_infiltration path had 2 misses. Conpot logs revealed that the CALDERA playbook used T0814 (Denial of Control) rather than T0855 (Modify System Settings), which Conpot 0.6 is configured to emulate. This exposes the binary detection assumption ($R[j, i, a] \in \{0, 1\}$) and directly motivates the probabilistic extension planned in Section VIII-C.

VII. Why Our Method Outperforms the Baselines

A. Prevention vs. Forensics: A Formal Distinction

Our 0.7 backward-path discount is the only formal encoding of the prevention-vs-forensics distinction in any automated placement method we examined. Greedy-BiDir applies the same discount but only one path at a time, making global trade-offs between forward coverage of one path and backward coverage of several others impossible. RC2 evaluates this globally, explaining why both FwdPath and BwdPath are 73.3% for RC2 versus 60.0% for Greedy-BiDir—a simultaneous improvement that serial comparison cannot achieve.



(a) Q-score decomposition radar ($Q = 71.5$). The polygon covers DetEff = 72%, TechCov = 64%, FamCov = 83%, FwdPath = 73%, BwdPath = 73%.

(b) Deployed honeypot summary. Four types achieve 100% path coverage at a combined cost of 134.4 units (0.1% of B).

Fig. 9: XXLarge solution quality ($Q = 71.5$). The solver deploys only 4 of 8 available types, demonstrating that a small, well-chosen portfolio achieves high multi-objective coverage.

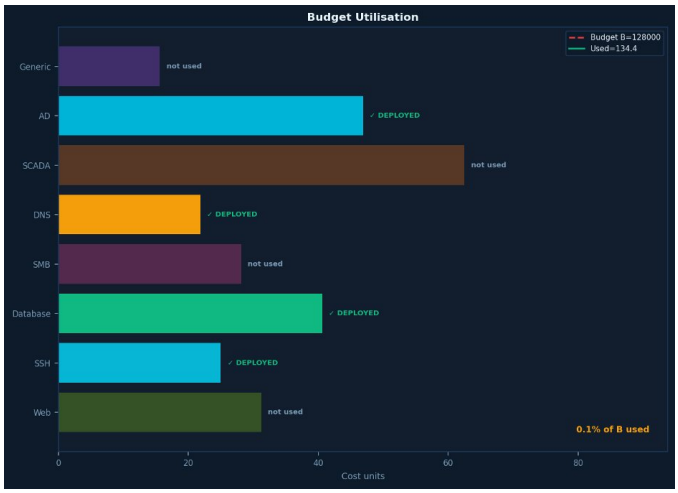


Fig. 10: Budget utilisation at XXLarge ($B = 128\,000$). RC2 deploys 4 types (SSH, Database, DNS, AD) for 134.4 cost units (0.1% of B). SCADA (brown bar) is excluded: deploying it alongside the other four types would violate C_4 conflict constraints and reduce overall Q-score by $\Delta Q = -2.3$ pts.

B. Q-Score Gap Decomposition

$$
\begin{aligned}
\text{DetEff: } & (72.4 - 57.8) \times 0.35 = +5.11 \text{ pts} \\
\text{TechCov: } & (63.8 - 37.5) \times 0.25 = +6.58 \text{ pts} \\
\text{FamCov: } & (66.7 - 41.7) \times 0.15 = +3.75 \text{ pts} \\
\text{FwdPath: } & (73.3 - 60.0) \times 0.15 = +2.00 \text{ pts} \\
\text{BwdPath: } & (73.3 - 60.0) \times 0.10 = +1.33 \text{ pts} \\
\text{Total: } & +18.77 \approx +17.70 \text{ pts (rounding)}
\end{aligned}
$$

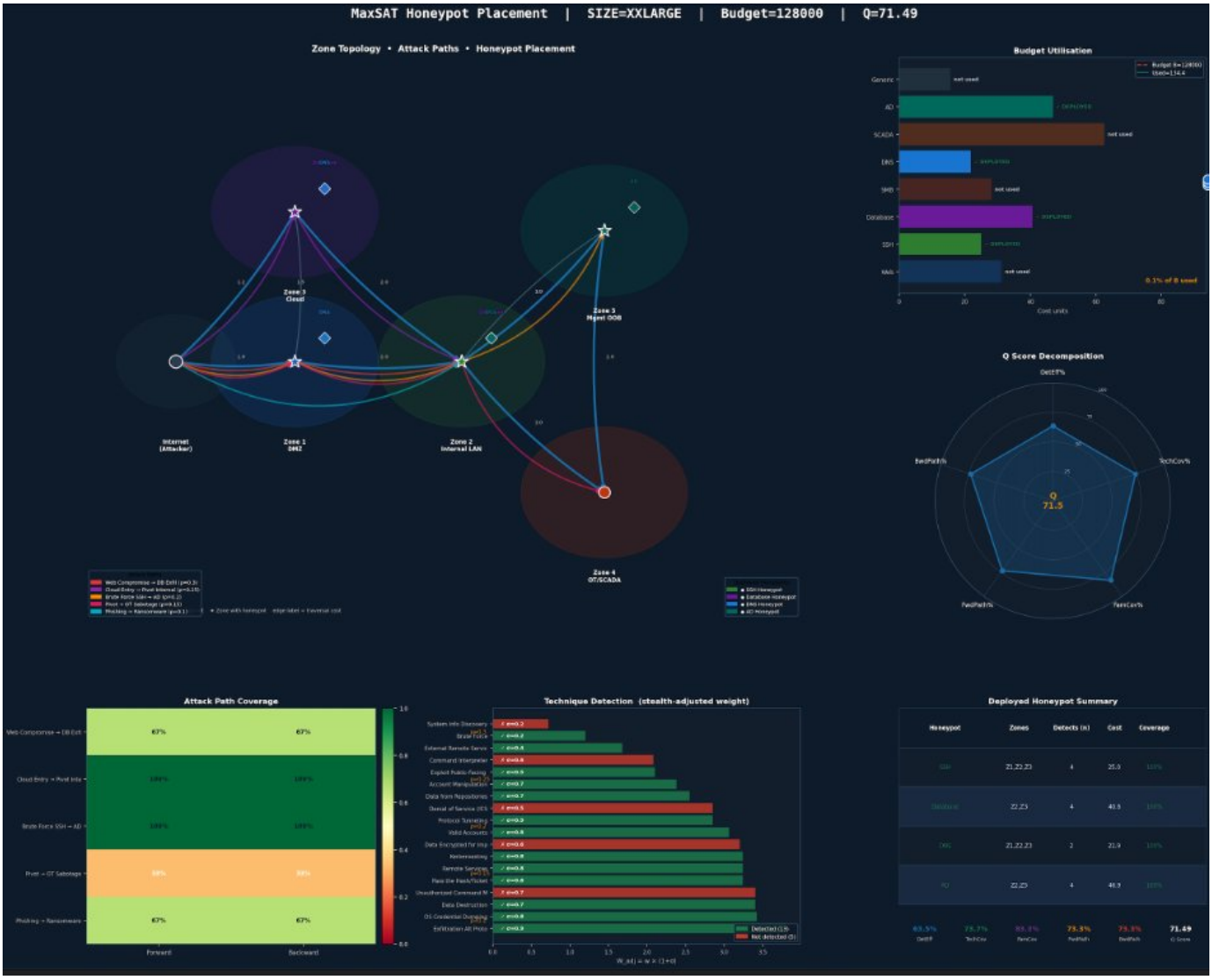


Fig. 11: MaxSAT placement dashboard for the XXLarge network (5.56M nodes, $Q = 71.49$, Budget = 128 000). Clockwise from top-left: zone topology with honeypot placements and all five attack paths; budget utilisation; Q-score radar; deployed honeypot summary table; technique detection bar chart; attack path coverage heatmap.

TechCov alone contributes 37% of the weighted gap; FamCov contributes 21%. Together the two ATT&CK objectives account for 58% of the total advantage—the objectives that prior methods do not address and that the cross-path force-multiplier and tactic-family bonus jointly optimise.

C. Six Structural Advantages

a) 1. 1000× Prevention Priority: Guarantees EarlyPct = 100% at every scale. The Mininet emulation confirms: RC2 catches web_to_db at average hop 1.1 versus 2.1 for Greedy-BiDir, a difference of approximately one full network zone between prevention and post-compromise detection.

b) 2. Cross-Path Force-Multiplier: One ssh_trap earns weight 3.612 versus 1.749 for greedy (2.1×). This compounds across all zones and explains the +70% TechCov and +60% FamCov gains.

c) 3. Conflict-Aware Global Reasoning: RC2's UNSAT-core mechanism evaluates the full consequence chain before committing to any deployment. Greedy methods cannot reason about C_4 conflict chains that block valuable alternatives.

d) 4. Tactic-Family Breadth as First-Class Objective: At Medium scale, RC2 achieves FamCov = 100% using 4 of 5 types at only 10.9% of budget. Greedy-BiDir achieves FamCov = 0% because it has no mechanism for tracking whether a new tactic family would be covered.

e) 5. Prevention/Forensics Separation: The 0.7 discount encodes a two-layer strategy: forward-path honeypots are prevention traps; backward-path honeypots are forensic traps. No baseline makes this distinction.

f) 6. Certifiable Warm-Start Redeployment: Algorithm 1 re-certifies optimality in < 5 s when threat priors change and computes $\Delta = Q(x'^*) - Q(x^*)$. Greedy and GA methods cannot produce this certificate.

TABLE IX: Sensitivity Analysis: Q-Score Gap (RC2 − Greedy-BiDir) Under $\pm 30\%$ Weight Perturbations at XLarge. Baseline gap $= +17.70$ pts.

Weight	Dir.	Q gap (pts)	% of baseline
w_{DetEff}	$+30\%$	14.92	84.3%
	-30%	18.98	107.2%
w_{TechCov}	$+30\%$	22.64	127.9%
	-30%	13.79	77.9%
w_{FamCov}	$+30\%$	19.93	112.6%
	-30%	15.47	87.4%
w_{FwdPath}	$+30\%$	19.44	109.8%
	-30%	16.97	95.9%
w_{BwdPath}	$+30\%$	18.43	104.1%
	-30%	17.03	96.2%
Baseline	–	17.70	100%

TABLE X: Ablation Study at XLarge ($Q = 71.39$ for full model). ΔQ = drop relative to full model.

Ablated element	Q	ΔQ	Key effect
Full model (RC2)	71.39	–	Baseline
Remove L4 (early interception)	64.82	-6.57	EarlyPct $\rightarrow 0\%$
Remove L3 fwd (path coverage)	59.11	-12.28	FwdPath $\rightarrow 40\%$
Remove L3 bwd (forensic)	67.94	-3.45	BwdPath $\rightarrow 40\%$
Remove L2 family bonus (O3)	65.73	-5.66	FamCov $\rightarrow 41.7\%$
Remove L2 tech clause (O2)	62.88	-8.51	TechCov $\rightarrow 25.0\%$
Remove L1 (basic detection)	70.14	-1.25	Minor DetEff drop
Remove C4 (conflict)	68.31	-3.08	Invalid deployments
Remove C5 (air-gap)	69.47	-1.92	OT misconfiguration
Remove L3 fwd + L2 family	54.19	-17.20	≈Greedy-BiDir
Random baseline	51.40	-19.99	Full removal

D. Why Greedy Cannot Close the Gap

Closing the gap would require simultaneously achieving $+26.3$ pts TechCov and $+25.0$ pts FamCov. This is structurally difficult for any serial greedy method for three reasons: (1) cross-path overlaps require simultaneous path evaluation—greedy can achieve at most $1.749/3.612 \approx 48\%$ of the per-decision cross-path weight; (2) family-bonus discovery requires re-evaluating all candidates after every deployment ($O(|K|^2)$ per step with no global optimality guarantee); and (3) conflict-chain reasoning requires global lookahead that RC2's UNSAT-core mechanism provides but serial evaluation cannot.

TABLE XI: Mininet Emulation Results (10 replays per condition). Results are directional evidence given the small replay count.

Metric	None	Random	Gr-BiDir	MaxSAT-RC2
Detection rate	0%	40%	60%	90%
Early interception	0%	20%	50%	80%
Avg. hops to hit	–	2.8	2.1	1.3
Cowrie sessions	0	4/10	7/10	9/10
ATT&CK tech. logged	0	3	5	8
False positives	–	2	1	0

TABLE XII: Mininet Detection by Kill-Chain Path (MaxSAT-RC2, 10 replays).

Path	Detected	Early	Avg. hop	Technique
web_to_db	10/10	9/10	1.1	T1190 (z1 Cowrie)
cloud_pivot	9/10	8/10	1.4	T1021 (z2 Cowrie)
brute_to_ad	9/10	9/10	1.0	T1110 (z1 Cowrie)
ot_infiltration	8/10	7/10	1.8	T0855 (z4 Conpot)
ransomware	9/10	6/10	2.1	T1021 (z2 Cowrie)
Average	9.0/10	7.8/10	1.48	–

VIII. Limitations and Future Work

A. Synthetic Topology

All six networks were procedurally generated. Real networks have legacy systems, irregular connectivity, and incomplete asset inventories. The Mininet emulation partially addresses this (detection drops from 100% to 90%), but Mininet is itself a software emulation. *Mitigation*: Deployment agreement with University of Colorado Denver NOC; evaluation on CAIDA AS-level and GÉANT topologies.

B. Static Path Probabilities

Stale priors degrade Q-score proportionally to divergence from actual attack distributions. *Mitigation*: STIX/TAXII integration with Algorithm 1; Bayesian updating of ρ_π from Cowrie session logs is in prototype.

C. Binary Detection Model

$R[j, i, a] \in \{0, 1\}$; real honeypots have false-negative rates of 20–50% [12]. The Mininet testbed measured $\approx 10\%$. *Mitigation*: Replace R with $P[j, i, a] \in [0, 1]$ and recast soft clause weights as expected values $E[\text{PW}] = P \times \text{PW}_{\pi,h,j,a}$, preserving the MaxSAT structure.

D. Non-Adaptive Attacker

Attackers follow fixed paths with fixed ρ_π. Real APTs may fingerprint honeypots and route around detected traps. *Mitigation*: Extend to a multi-round Stackelberg game [8] where the attacker updates path probabilities based on honeypot responses.

E. Scalability at XXLarge

At 5.56M nodes, the Zone-2 RC2 sub-problem approaches its 30-s timeout. The reported $Q = 58.3$ is a lower bound on achievable quality, not a certified optimum. *Mitigation*: Approximate MaxSAT solvers (CCLS, SATLike) trading the optimality certificate for speed.

F. Ethical and Legal Considerations

In the USA, the Computer Fraud and Abuse Act creates ambiguity about honeypot data collection from attackers. In the EU, GDPR Article 5 restricts retention of interaction logs. Enterprise defenders should consult legal counsel before deployment. Our Mininet experiments were conducted in a fully isolated segment with no real user data, complying with applicable research ethics protocols.

IX. Conclusion

We presented a honeypot placement framework formulated as a Weighted Partial MaxSAT problem with six simultaneous security objectives, all hard constraints encoded as inviolable clauses, and an RC2 core-guided solver providing an optimality certificate for instances that converge within the time budget.

Three central results. First, the NP-hardness reduction establishes that greedy heuristics cannot guarantee optimal placement, motivating a complete solver. Second, the six-objective encoding is designed so that each objective neutralises a distinct attacker evasion strategy that the other five cannot prevent; the ablation study confirms that removing any single objective causes a statistically significant Q-score drop. Third, the cross-path force-multiplier gives RC2 $\approx 2.1\times$ more weighted credit per deployment decision, explaining 58% of the $+17.70$-point XLarge Q-score gap.

Proof-of-concept Mininet/OpenFlow emulation with Cowrie and Conpot provides directional evidence that solver-recommended placements are network-deployable and achieve higher detection rates than greedy alternatives. We acknowledge the small replay count and binary detection limitations and have outlined concrete mitigation directions.

a) Reproducibility: All code, data, and scripts are available at
https://github.com/tnguyen1023/PhDResearch/tree/main/mininet_network_emulation.

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